Options Signal Engine

V2.1 FINAL

Manual execution · 8 assets · 2–5 DTE Rolling Fridays
Finnhub WS + yfinance · $0/month data

Hard halt / blocked | Data / gates | Approach context | Dynamic baselines (ATR/rolling avg) | Confirmed signal / output | Computed metrics | Confidence scoring

DATA LAYER

Finnhub WebSocket live ticks

8 assets × live ticks

SPY QQQ AAPL NVDA TSLA MSFT META AMZN

VIX poll	every 30s
Eco calendar	macro events
WS stale detect	>30s → halt
Resync	7-step protocol

yfinance bars + calendar

1m bars	cap 500
5m bars	cap 2000
15m bars	cap 1000
Daily	cap 252
Earnings	ticker.calendar
Holidays	2026 hardcoded

Computed Metrics

CVD	asyncio.Lock/asset
VWAP	intraday reset
ATR	14-period
Vol profile	VAH/VAL/POC
IV estimate	VIX × beta
OR high/low	9:30–10:00

Beta values

SPY=1.0 QQQ=1.15 AAPL=1.2
NVDA=1.8 TSLA=2.0 MSFT=1.1
META=1.3 AMZN=1.2

State Persistence SQLite

HOT MEM	Tick CVD, live candles, active detections
60S FLUSH	CVD snapshots, day context
5M FLUSH	Vol profile, LevelState, signal history, last ticks

WS Resync (7-step)

1. Detect stale (>30s)
2. Record gap_start / gap_end
3. Fetch missing 1m from yfinance
4. Reconstruct CVD → ESTIMATED flag
5. Rebuild volume profile
6. UNCERTAIN trackers → threshold=75
7. Quarantine: MEDIUM only 2min

GATE SEQUENCE (0 → 8)

Pre-Signal Checks

0 Macro event halt
FOMC/Fed: ±30min · CPI/NFP/PCE: ±15min
Hardcoded 2026 dates + Finnhub calendar backup
FAIL → hard halt, no signal

E Earnings check
ticker.calendar — within hold period?
yfinance. SKIP if earnings overlap with max hold window.

H Holiday check
Market open today? Early close?
Early close days: cutoff 12:30 PM

Core Gate Sequence

1 Location
abs(price - level) ≤ ATR × 0.3
Dynamic proximity. Solves TSLA/NVDA trap.

2 Approach context
5-candle classifier → 5 types
Priority: AGGRESSIVE→ABSORPTION→EXHAUSTION→
NEUTRAL allowed: confidence -15p

3 Volume signature
vol ≥ 1.5 × rolling_avg(last_10_bars)
Solves U-curve bias. Setup 3A uses inverse: vol < 0.8×.

4 Price signature
Consistent with approach context + setup
Displacement ratio, wick/body ratio, candle color

5 CVD signature
abs(cvd_turn) ≥ 2.0 × rolling_avg(last_10_cvd_turns)
OR CVD divergence present (+10 conf bonus). Solves order flow distortion.
CVD quarantine: cap contribution at 8pts

DETECTION ENGINE

Setups

S1 Liquidity Grab 1m

APPROACH	Aggressive
SESSION	Any (after 10:00)
TRIGGER	Sweep candle: wick > 0.3×ATR past level, body closes back. Displacement ratio > 0.6.
VOLUME	Rank top 25%
CVD	Turn > 2×avg
ENTRY	FVG midpoint (preferred). If no FVG: next candle open.
FVG	Gap between candle[-3].hi and candle[-1].lo

S2 OB Defense 5m+1m

APPROACH	Absorption
SESSION	After 10:00. TREND days only.
DAY BIAS	Bullish locked → LONG only
OB DEF	Last opposite-colored candle before displacement. Body > ATR×1.0, vol top 20%.
OB VALID	Formed within 60min. Not visited 2+ times since formed.
DEFENSE	candle.low to zone (±0.15%) slope: declining→fla

S3 Failed Auction VAR + Major

VERSION A — VAR

SESSION	After 11:00 only
CONDITION	Price outside VAH/VAL. Body closes back inside.
VOLUME	vol < 0.8× rolling avg (low-vol confirm)
CVD	Turn > 1.5×avg
TARGET	ALWAYS POC

VERSION B — MAJOR LEVEL

REQUIRE	Level score ≥ 8. Exhaustion approach.
WICK	wick/body > 2.5. Vol top 25%.
CVD	Turn > 2×avg

S4 Failed Breakout reversal

TRIGGER	After S2 failure. Retest closes wrong side.
CONFIRM	Next candle confirms direction + CVD > 1.5×avg
DIRECTION	Opposite of original S2

Approach Context Classifier 5-candle window

AGGRESSIVE PUSH	Priority 1
ABSORPTION	Priority 2

EXECUTION LAYER

2–5 DTE Rolling Fridays

Mon	Current Friday	
Tue	Current Friday	
Wed	Current Friday	
Thu	Current Friday	before noon
Thu	SKIP	after 1:00 PM
Fri	NEXT Friday	before noon
Fri	SKIP	after noon

Strike	ATM always
IV est	VIX × asset_beta
VIX >30	debit spread only
VIX ≥35	hard block
Max hold	5 trading days
Exit by	Day 3 if TP1 not hit

Spread check rule

ask ≤ est_premium + $0.20 → buy
ask > est_premium + $0.20 → skip
price drift >0.5% from signal → skip

Session Quality Gates

9:30–10:00	Hard block
10:00–11:00	Power hour
11:00–12:00	Mid morning
12:00–14:00	Dead zone
14:00–15:15	Afternoon
15:15–15:30	Power close
15:30+	Hard cutoff

Dead zone: score ≥8 only, no VAR.
Power close: score ≥9 only.

Day Bias locked at 10:00 AM

Day type (direction changes in last 30 OR bars)

≤4 changes	TREND DAY → S2 valid
5–9 changes	RANGE DAY
≥10 changes	RANGE DAY

Directional bias score

Daily trend bullish	+1
15m trend bullish	+1
Broke above ORH	+2
Price above VWAP	+1
Broke below ORL	-2
Price below VWAP	-1

≥+3	BULLISH locked → LONG only

6 Confidence score

Score ≥ 50 → MEDIUM · Score ≥ 75 → HIGH

0–100 composite. NEUTRAL bias day: threshold raised to 70.

7 Confirmation candle

Next candle closes in signal direction

WAIT if not yet. Max 2 agent calls per tracker. Skip if super-candle.

Super-candle: vol top 10% + body > ATR + CVD > 4×

8 Hard system rules

VIX, session window, cooldowns, budget, trend align

10 rules — any single FAIL blocks signal

≥1 of last 2 candles: body > ATR×1.0
AND that candle vol ≥ 2.0× rolling_avg_10
AND moving TOWARD level
AND abs(close[-1]-close[-3]) > ATR×0.8

max(high[-4:]) - min(low[-4:]) < ATR×0.8
≥3 of last 4 bars: vol > rolling_avg
AND avg_vol(last 4) > 1.2× rolling_avg
AND CVD flat or rising despite flat price

EXHAUSTION Priority 3

body[-1] < body[-2] < body[-3]
AND all 3 bodies < ATR×0.4
AND avg_vol(last 3) ≤ avg_vol(-4 to -6)×1.1
NO wick requirement (event candle, not approach)

MOMENTUM Priority 4

≥4 of last 5: close[i] > close[i-1]
AND each body > ATR×0.15 (no doji)
AND total progress abs(close[-1]-close[-5]) > ATR×0.5

NEUTRAL Default fallback

None of the above match. Detection not blocked. Confidence contribution = 0pts (max -15 from potential).

Confidence Scoring 0–100 composite

COMPONENT	MAX	KEY THRESHOLDS
Location quality	20	POC/VAH/VAL=20, PDH/PDL=15, ORH/ORL=10, weekly/monthly=18, confluence=+10 bonus
Volume signature	20	ratio≥2.0=20, ≥1.5=12, ≥1.2=5, <1.2=0 (blocks)
Price signature	20	displacement>1.0=20, 0.7–1.0=14, 0.4–0.7=8, wick>0.5ATR=+5 bonus
CVD signature	25	turn>4×=25, 2–4×=15, 1–2×=8, divergence=+10 bonus, flat=0. Quarantine cap: 8pts
Approach context	15	exhaustion/absorption=15, aggressive=12, momentum=10, neutral=0

Thresholds

≥75 HIGH 50–74 MEDIUM <50 NO SIGNAL

NEUTRAL bias day: min 70 Quarantine max: 63

CVD — 4 Metrics

1. Turn magnitude

abs(cvd_event) / avg_abs_cvd_per_candle (rolling 10 bars)

≥2× = required minimum · ≥4× = strongly institutional

2. Divergence

Price new highs + CVD lower highs = distribution. Price new lows + CVD higher lows = accumulation.

+10 confidence bonus when detected

3. Slope at level

CVD declining before level → flattens/turns AT level = absorption moment

4. Sustained vs spike

Turned and STAYED turned 2+ candles = real institutional positioning

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≤-3	BEARISH locked → SHORT only
-2 to +2	NEUTRAL · both · threshold 70

Override: level score ≥9 → counter-bias allowed

Level System

Min score	≥7 (required)
Target per asset	6–8 levels
Confluence	within 0.30% → max+2 (cap 12)

Volume interaction multiplier

0 interactions	×0.5
1 interaction	×1.0
2 interactions	×1.3
3+ interactions	×1.5

10 Hard Rules

1. No signals before 10:00 AM ET
2. No signals after 3:15 PM ET
3. No signals during Tier-1 macro events
4. No signals if earnings within hold period
5. No signals when VIX ≥ 35
6. No counter-trend (unless level score ≥9)
7. No duplicate signals same level same day
8. Max 3 signals per asset per day
9. Minimum RR 2.5:1
10. One agent at a time (asyncio.Lock)

Signal Output → Telegram

CONFIRMED SIGNAL BRIEF

Asset + direction	NVDA LONG
Setup type	S1 Liquidity Grab
Level	PDH $875.40 (score 9)
Entry	$875.20 (FVG mid)
Contract	875C Fri expiry
IV estimate	~38% (approx)
Confidence	82 HIGH
Session	Power Hour 5/5
Day bias	BULLISH (+4)
Spread check	ask ≤ est + $0.20
TP1 / TP2 / SL	auto-computed 2.5:1
Max hold	Day 3 exit

WAIT signal: no Telegram. Tracker continues. Max 2 agent calls per tracker.

MODULE FILE MAP — V2.1 BUILD TARGETS

New files

- detection/liquidity_grab.py
- detection/defense.py
- detection/failed_auction.py
- detection/approach.py
- detection/metrics.py
- detection/confidence.py

Modify — detection

- detection/failed_breakout.py
- detection/level_state.py
  - → UNCERTAIN state
  - → safe iteration
  - → WAIT tracker logic
- levels/builder.py

Modify — core / context

- core/multi_engine.py
  - → route new setups
  - → asyncio.Lock
- core/gates.py
  - → Gate 0, earnings
  - → ATR×0.3 proximity

Modify — agent / data

- agent/agent.py
  - → asyncio.Lock
- agent/brief.py
  - → macro, earnings, IV
- agent/tools.py
  - → 2000 char truncation

data/persistence.py
data/calendar.py

→ vol confirm multi
→ score ≥7 threshold

context/options_context.py
context/day_context.py

data/cvd_engine.py
data/candle_store.py

Options Signal Engine v2.1 — Design locked. Manual execution. $0/month data.

Philosophy: "Are institutions actively positioning at this price level right now?"